UK Health Security Agency

A Field Services minimum standard for invasive GAS cluster detection in England

Standard Operating Procedure

Background

Consistent Field Services iGAS cluster detection is needed to support HPT cluster detection and enable timely and proportionate public health responses. Field Services cluster detection should not replace HPT context surveillance of iGAS cases, but rather complement and provide additional safety-netting to improve sensitivity and timeliness of cluster detection. This SOP describes suggested steps to detect residential, community setting and social risk factor clusters of iGAS cases using the iGAS dashboard and the extractable case-level data used by the dashboard. The SOP also provides a set of audit standards to ensure consistency across the Field Services and allow teams to assess the implementation of the cluster review process as part of each team's quality improvement cycle.

Cluster Definitions

The following cluster definitions are for Field Service cluster surveillance. These should not replace HPT context surveillance of iGAS cases, and HPTs should take appropriate public health action if they become aware of any iGAS clusters within their regions/areas that may not fulfil the criteria below.

Cluster	GAS/iGAS Cases		Spatial Link		Temporal Link		Epidemiological Link		Microbiological Link
Residential cluster	• iGAS cases (sterile specimen) • iGAS cases (non-sterile specimen) • Non-invasive GAS cases (non-sterile specimen)	***AND***	Matching residential post code	***AND***	6 months (rolling) between isolate specimen dates	***AND***	Same residence (e.g. household, care home, custodial setting)	***AND***	Not dependant on emm-type
Community setting cluster	• iGAS cases (sterile specimen) • iGAS cases (non-sterile specimen)	***AND***	Cases reside within the same Upper Tier Local Authority	***AND***	6 months (rolling) between isolate specimen dates	***AND***	A shared primary **AND** secondary **AND** Tertiary context If tertiary context is not available, then match on secondary context ***OR*** Shared exposure(s) in the 7 days prior to onset of iGAS	***AND***	Matching emm-type ***OR*** Not dependant on emm-type if epidemiological link is community health services delivered in the home **and** emm-typing is not yet available.
Social Risk Factor cluster	• iGAS cases (sterile specimen) • iGAS cases (non-sterile specimen)	***AND***	Cases reside within the same Upper Tier Local Authority	***AND***	6 months (rolling) between isolate specimen dates	***AND***	Shared social risk factor	***AND***	Matching emm-type

Surveillance Aims and Objectives

Aim

To detect iGAS clusters within regions of England, to enable rapid implementation of public health measures.

Objectives

1. To identify new clusters of iGAS cases.
2. To identify and assess new iGAS cases that may be linked to established iGAS clusters.
3. To escalate newly identified clusters, or clustered cases, to HPTs for timely assessment and public health action where appropriate.

Overview

We recommend the cluster detection process is conducted weekly using the following steps.

- Step 1 - Detect new cases within existing clusters

  Purpose: To meet surveillance objective 2, identify new iGAS cases linked to established iGAS clusters, using cluster definitions.

- Step 2 - Detect new residential, community settings and social risk factor clusters

  Purpose: To meet surveillance objective 1 - to identify new clusters of iGAS cases.

- Step 3 - Broad review for potential new clusters: Cases with missing data needed to establish epidemiological links.

  Purpose: This step groups cases using spatial, temporal, and microbiological links only (without considering other epidemiological links). The rationale behind this step is that epidemiological links (e.g. primary contexts or exposures within 7 days) are not consistently recorded within the correct fields in CIMS and so may be missing from data extracted from the iGAS linelist in Health Protection Datawarehouse (HPDW).

- Step 4 - Communicate findings (including any null findings) from the cluster review process to HPTs as per locally agreed arrangements to ensure the appropriate documentation in CIMS.

  Purpose: For health protection work including further risk assessment, information gathering and public health action if relevant.

Optional steps

- Step 1 – Update cluster caches with CIMS incident numbers for any clusters where the HPT has declared an incident with pending public health action e.g. convening an incident management team (IMT)

  Purpose: Such that any new cases that are part of existing clusters deemed to require public health action can be identified and communicated to HPTs (as part of Step 3).

Access requirements

iGAS Dashboard – Front-end

To access the dashboard, you need to have previously applied to join the relevant region-based security user group by completing an access form.

Regions Data Science - Dashboards Access Request

iGAS linelist in team workspaces

A case-level dataset of iGAS cases from the past 12-months (for all regions) is available as a view in each Field Service Team database, in HPDW. To gain access to the database, users must first obtain permission from LAMa by selecting the relevant Field Service team project/database.

CIMS

Access to the CIMS front-end to access individual case records may be useful for any additional investigations to further establish any epidemiological links between cases by reviewing case notes.

Full details of all requirements to create a CIMS account and access training materials and guidance can be found in the CIMS User Area. Staff will need to have completed all mandatory training, particularly information governance training, before requesting access.

Site addresses:

CIMS – https://prod.cims.ukhsa.gov.uk/maven/main.do
iGAS Dashboard - https://igas-dashboard.apps.ocp-az-uks.ukhsa.gov.uk

Local Cluster Cache

This SOP assumes teams keep a cluster cache, which records details of cases grouped by cluster type (residential, community settings and social risk factor). The purpose of a cache is to facilitate both automated, and/or manual, cluster detection processes so as to prevent the recurrent detection of clusters found in prior weeks. Each case should have a unique identifier (preferably the CIMS identifier) and each cluster should be labelled with a unique cluster identifier (cluster ID) – i.e. each case within a cluster should be labelled with the same cluster ID. Whilst teams are free to decide the structure of their caches, the following variables structure would be recommended. Separate tabs for residential, community setting, and social risk factor clusters are recommended, which will allow for cross checking of cases that appear within more than one type of cluster e.g. a residential cluster in a nursing home, with a link to a district nursing cluster.

Field	**Reason for inclusion**
Case identifier	Unique case level identifier. Can be process generated or the CIMS identifier
Sex	PII. Field available if teams extract data from the iGAS linelist saved in HPDW.
Age	Additional patient information
Emm type	Cluster detection – broad review, community settings and social risk factor
Invasive indicator	Identify iGAS or GAS case
Date of birth	PII. Field available if teams extract data from the iGAS linelist saved in HPDW.
Specimen date	Cluster detection – All
Address	Number or building, street and locality concatenated into one field. Can be used to check whether cases with the same residential post code have the same address.
Postcode	Cluster detection – residential settings
UTLA	Cluster detection – broad review, community settings and social risk factor
LTLA/LA	Cluster detection – broad review
Primary context	Cluster detection – community settings
Secondary context	Cluster detection – community settings
Tertiary context	Cluster detection – community settings
Context name	Cluster detection – community settings
Date entered	To identify newly reported cases of iGAS
Risk factor	To include: People Who Inject Drugs (PWID) People Experiencing Homelessness (PEH)

	Prison or Prescribed Place of Detention (PPD) settings
Exposures in the 7 days prior to onset of iGAS	This can include: Hospital or healthcare exposures Community health services delivered at home (CHSDH) Social care support Care home PWID Homeless Prison or Prescribed Place of Detention (PPD) settings Nursery/school setting Travel
Reviewer	Initials of FS staff member who conducted the cluster detection process.
Review date	Date when the cluster detection process was conducted.
Cluster ID	Unique ID for the cluster
Cluster type	Residential, community settings and social risk factor clusters
Cluster detection date	Date the cluster was first detected. Should be the same as the review date when the cluster was first detected.
CIMS incident identifier	Incident number assigned to cluster if an incident is declared, and incident record is created on CIMS.
Linked cluster ID	Optional field to identify cases part of a different cluster e.g. a residential cluster (care home) linked by a case to community setting cluster

An additional triaging potential clusters cache can also be used for Step 3 in the process.

Case level data – line lists

To detect new residential, community settings and social risk factor clusters, linelists of iGAS and GAS cases with specimen dates in the previous 6 months (as a minimum) will be required. This linelist can then be compared against a cache of cases grouped into previously identified clusters to ensure that the same clusters are not continuously detected. Case level data of iGAS infections can be accessed via the iGAS linelist view in all Field Service databases in HPDW.

The CSV linelist generator from the dashboard front-end is a quick and intuitive way to extract a dataset of iGAS cases using point-and-click**. However, it does not allow teams to be able to detect clinically diagnosed iGAS cases with non-sterile specimens (included in all cluster definitions) and pertinent non-invasive GAS cases (included in residential cluster definition). The dashboard front-end does not provide the fields required to exclude other non-invasive GAS cases (e.g. scarlet fever) that are not relevant for cluster detection which would introduce unnecessary noise.**

iGAS linelist view in all Field Service databases

1. Open Microsoft SQL Server and select the server name: SQLCLUSCOLHPO19\HPO19.
2. Navigate to the relevant Field Services Team database.
3. Using SQL queries to extract case level data from the iGAS linelist (igas_linelist) view saved in each teams database space.
    a. **To ensure clinically diagnosed iGAS cases with non-sterile specimens (that are labelled as non-invasive GAS) and relevant non-invasive GAS cases are included within the line list, ensure that all GAS cases are extracted i.e. do not filter using the invasive indicator at the point of data extraction.**
4. Once the case line list has been extracted, exclude cases where the Invasive_Indicator is NULL or missing. Cases with ‘Y’ or ‘N’ should be included.
5. Name the extract and include a date. Then save to a secure drive suitable for files containing PII and either overwrite last week’s extract or save within a specific folder with all extracts.

Alternatively, the iGAS linelist view can be accessed via programming software (R, Python, STATA) and an ODBC link using relevant packages or drivers. In Field Services, this is commonly done using R and the odbc and DBI packages to extract data from tables or views within SQL databases.

iGAS Cluster Detection

Step 1: Detecting new cases that are part of existing clusters

1. Extract an up-to-date line list of iGAS (and GAS) cases using the options specified.
2. Identify any newly reported cases within the region.
    a. Newly reported cases are those reported since the last time the cluster detection process was conducted.
    b. Newly reported cases will be those with a date, in the date entered field, that is within ≤7 days from the current date.
3. Compare newly reported cases against existing residential clusters AND community setting clusters linked by community health services delivered in the home.
    a. If any newly reported cases match the post code of a previously identified residential cluster, and the specimen date is within ≤6 months of the most recent case within the previously identified cluster, then that case is part of an existing cluster.
    b. Note that newly reported cases will likely have non-typeable in the emm-type field (as emm-typing will be pending) and so would only be able to fulfil the criteria for residential clusters OR community setting clusters linked by community health services delivered in the home.
4. Add the details of new case(s) within residential clusters to the residential cluster tab in the cluster cache and add the same cluster ID as the cluster it belongs to.
5. Compare the entire line list (including newly reported and untyped cases) against the cluster cache to identify newly typed cases part of existing community settings or social risk factor clusters.
    a. Newly typed cases will only have been compared to residential clusters and community setting clusters linked by community health services delivered in the home until this point.
6. Add the details of new case(s) within community settings or social risk factor clusters to the cluster cache and add the same cluster ID(s) as the cluster(s) they are a part of.
7. Check for clusters with cases in common (both intra- and inter-cluster type) by identifying CIMS identifiers that appear more than once.
8. Cross-matching and add cluster IDs of linked clusters to the Linked Cluster ID column in the cache for cases that appear in more than one type of cluster.
9. Add Linked Cluster ID columns if needed.

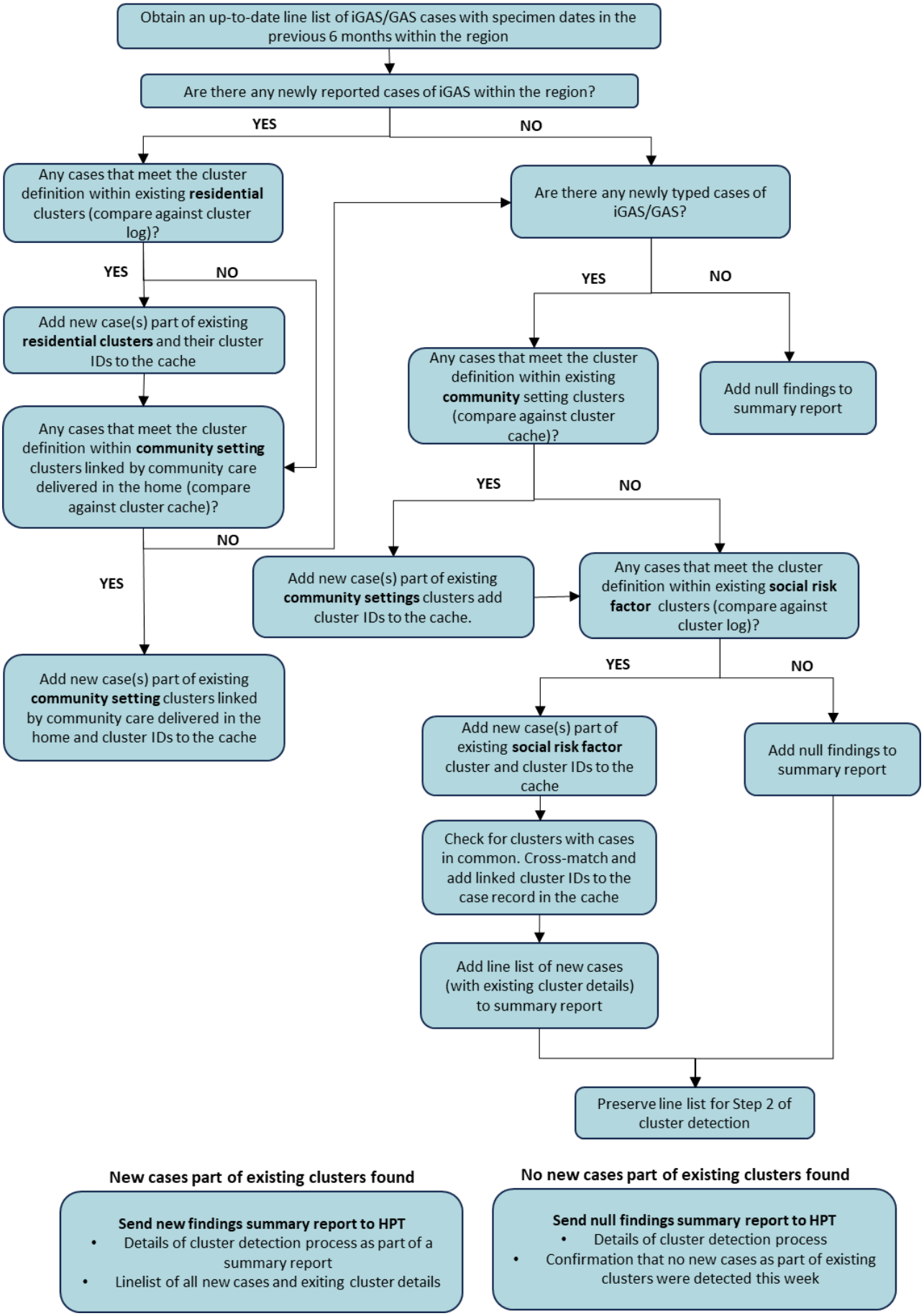

Figure 1 Algorithm for Step 1 - to identify new cases part of existing clusters

Step 2: Detecting new residential, community settings and social risk factor clusters

1. Using the full line list to identify clusters of two or more cases that meet the following definitions:
    a. **Residential clusters:** same residential post code, specimen dates within ≤6 months between each case (rather than between earliest and latest case).
    b. **Community setting clusters:** same UTLA, specimen dates with ≤6 months between each case, the same primary and secondary context, OR the same exposure within 7 days (excluding social risk factor).
        i. For cases linked due to district nursing or other community health services delivered in the home, detection should not depend on emm-type if cases meet the spatial, temporal, and epidemiological links AND emm-typing results are unavailable.
    c. **Social risk factor clusters:** same UTLA, specimen dates with ≤6 months between each case, the same social risk factor
    d. Cases should be able to meet several cluster definitions (i.e. are part of more than one cluster).
2. Exclude any clusters previously detected.
    a. These can be identified by excluding cases that already appear in the cluster cache either by comparing the CIMS identifier or CIMS identifier and cluster ID.
    b. This prevents the same clusters being detected week on week.
3. After excluding already detected clusters, any sets of two or more cases that meet the cluster definitions will be new clusters.
    a. These cases can then be saved in an intermediate/staging cache if teams wish to conduct any additional assessment of epidemiological links by reviewing CIMS case records.
4. New clusters should be saved in the cluster cache, with a specified cluster type (residential, community settings or social risk factor) and a new unique cluster ID.
5. Check for clusters with cases in common (both intra- and inter-cluster type) by identifying CIMS identifiers that appear more than once.
6. Cross-matching and add cluster IDs of linked clusters to the Linked Cluster ID column in the cache for cases that appear in more than one type of cluster.
    a. Add additional Linked Cluster ID columns as needed.

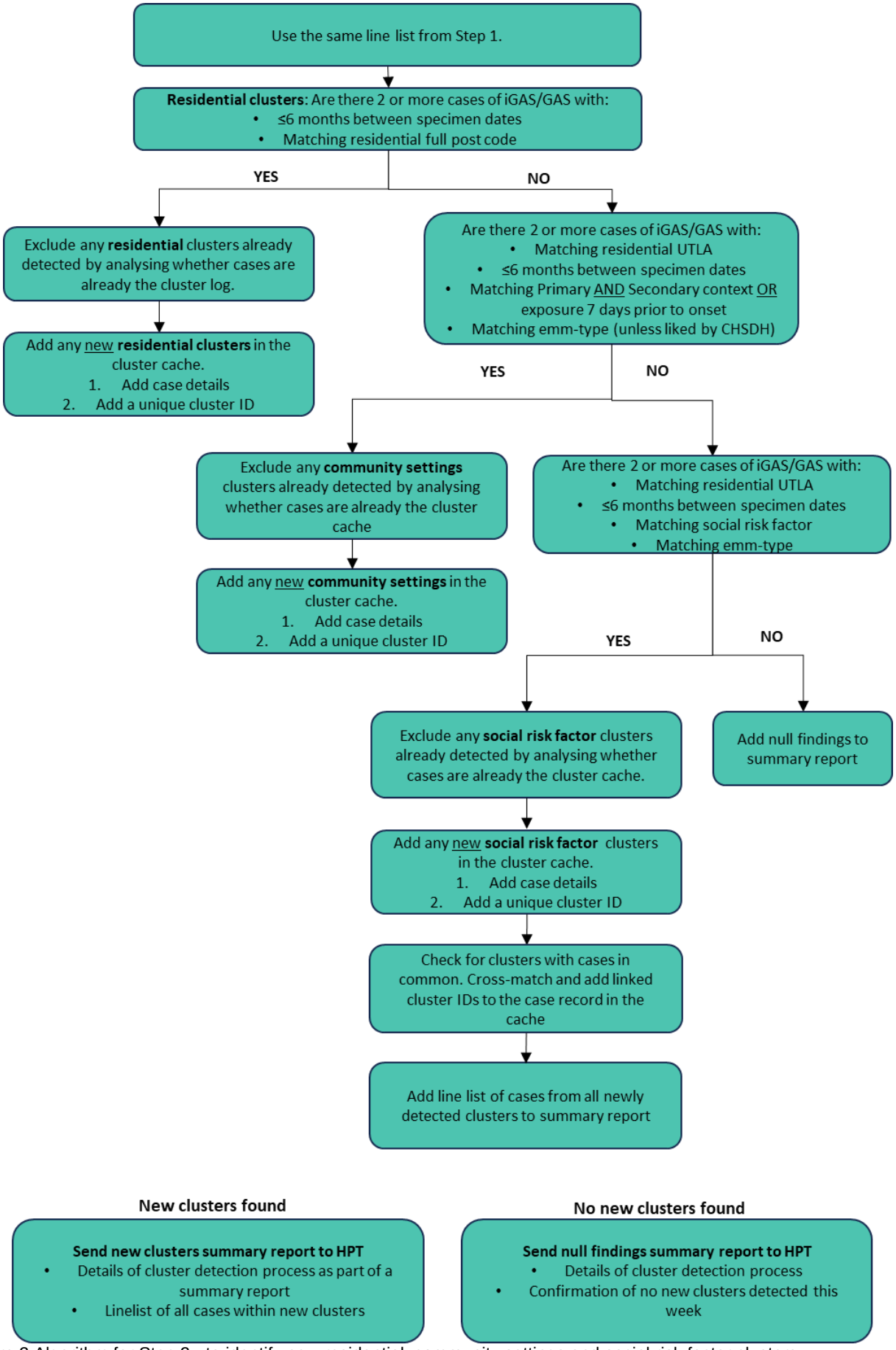

Figure 2 Algorithm for Step 2 - to identify new residential, community settings and social risk factor clusters

Step 3: Broad review for potential new clusters

1. Use the same linelist from step 1.
2. Identify any clusters of two or more cases that have:
    a. Same LTLA, same emm-type and ≤6 months between each case
3. Remove any cases or clusters already detected in parts 1 or 2.
4. This should leave a list of cases with potential epidemiological links that can be reviewed by HPTs.

Step 4: Reporting findings of cluster detection process

Share findings from steps 1 to 3 with the relevant HPT according to agreed local approaches on how new cases and clusters are reported. FS consultants or senior epidemiologists should review the findings and give final approval.

At a minimum, HPTs should a line list as part of a weekly summary report, which should include:

- New cases that are part of existing clusters (include the CIMS incident number for each cluster).
- Cases that form new clusters, for the HPT to conduct a risk assessment.

Instances where no clusters are detected should also be documented in the weekly summary report. To provide context, teams should also include a brief overview of the methods used to identify clusters (e.g., cluster definitions).

Optional

Weekly update cluster cache

For future audits and evaluations, it may be useful to regularly review whether clusters detected by FS processes and referred to HPTs resulted in any public health action or incident response. CIMS can be checked to ascertain whether the cluster has been escalated to an incident with a CIMS incident number. If so, add the CIMS incident number to all cases within the same cluster.

Auditable Standards

The following auditable standards apply to field services teams.

- The FS team iGAS detection process has been implemented to be able to detect residential, community settings and social risk factor clusters (at a minimum) as per cluster definitions.
- FS teams and HPT have an agreed local arrangement for the communication of outputs from cluster detection, as well as documentation of this in CIMS.
- FS teams maintain a cache of detected iGAS clusters.
- FS have identified staff who undertake iGAS cluster detection on a weekly basis.
- FS have identified senior staff to review findings from the weekly iGAS cluster detection process.
- The FS cluster cache allows teams to record evidence of when the last assessment was conducted.
- The FS cluster cache allows teams to record evidence of who conducted each weekly detection process.
- The FS cluster cache allows teams to record evidence of cluster review outcomes.
- FS teams send weekly summaries of findings from the cluster detection process to HPTs, including reports of null findings.